Blessed G/Michael Catholic Secondary School

Semester 1 Chemistry Final Examination for Grade12

Academic Year: 2024 / 2018 E.C. Time Allowed: 2:00'

Name: _____ **Grade & Sec:** Grade12 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** A weak acid HA has a Ka = 1×10^{-5}. Which change will increase the pH of its aqueous solution the most?

A. Diluting the solution **B.** Adding NaCl

C. Adding HCl **D.** Lowering the temperature

____ **2.** Which of the following pairs will form a buffer solution?

A. HCl and NaCl **B.** NH4Cl and NH4OH

C. NaOH and NaCl **D.** H2SO4 and NaSO4

____ **3.** The conjugate base of an acid is strongest when the acid has:

A. Large Ka value **B.** Small Ka value

C. High concentration **D.** Low molecular mass

____ **4.** Which statement about a buffer solution is correct?

A. Its pH does not change on adding strong acid

B. It completely neutralizes added acid

C. It resists change in pH on addition of small amounts of acid or base

D. It always has pH = 7

____ **5.** Which of the following salts will produce a basic solution in water?

A. NH4Cl **B.** Na2CO3 **C.** NaCl **D.** KNO3

____ **6.** Which solution has the highest pH?

A. 0.1 M HCl **B.** 0.1 M NaOH **C.** 0.1 M CH3COOH **D.** 0.1 M NH4Cl

____ **7.** Which of the following correctly represents the relationship between Ka and Kb of a conjugate acid–base pair at 25°C?

A. Ka + Kb **B.** Ka × Kb = Kw **C.** Ka - Kb = Kw **D.** Ka / Kb = Kw

____ **8.** At 25°C, the pH of pure water is 7. If temperature is increased, the pH of pure water will:

A. Increase **B.** Decrease **C.** Remain 7 **D.** Become zero

____ **9.** A weak acid has pKa = 4. If the ratio of salt to acid is 100, the pH of the buffer is:

A. 2 **B.** 4 **C.** 6 **D.** 8

____ **10.** During the titration of a weak acid with a strong base, the pH at half-equivalent point is equal to:

A. 7 **B.** pKa of the acid

C. pKb of the base **D.** 14 - pKa

____ **11.** 20 mL of H2SO4 solution requires 40 mL of 0.5 M NaOH for complete neutralization. The molarity of H2SO4 is:

A. 0.25 M **B.** 0.5 M **C.** 1.0 M **D.** 2.0 M

____ **12.** Given:

I. $Cu^{2+} + 2e^- \rightarrow Cu$, $E° = 0.34$ V

II. $Cu^{2+} + e^- \rightarrow Cu^+$, $E° = 0.15$ V

What is the electrode potential E° for the reaction $Cu^+ + e^- \rightarrow Cu$?

A. 0.53 V **B.** 0.38 V **C.** 0.3 V **D.** 0.9 V

____ **13.** According to Faraday's second law, mass deposited is proportional to:

A. Atomic mass **B.** Equivalent mass

C. Density **D.** Molarity

____ **14.** The electrode potentials of half-cell reactions are:
$Zn^{2+} + 2e^- \rightarrow Zn$, $E° = -0.76$ V
$Fe^{2+} + 2e^- \rightarrow Fe$, $E° = -0.41$ V (reduction)
The emf of the cell $Fe^{2+} + Zn \rightarrow Zn^{2+} + Fe$ is:

A. +0.35 V **B.** -1.17 V **C.** -0.35 V **D.** +1.17 V

____ **15.** The standard reduction potential of metals A, B, C and D are -0.40 V, +0.34 V, +0.8 V and -1.67 V. The metal that can displace all others is:

A. D **B.** C **C.** A **D.** B

____ **16.** The standard emf of a galvanic cell involving cell reaction with n = 2 is found to be 0.295 V at 25°C. The equilibrium constant of the reaction would be:

A. 1.0×10^{10} **B.** 2.0×10^{11} **C.** 4.0×10^{12} **D.** 1.0×10^{2}

____ **17.** Which response contains all the following statements that are correct, and no others?
I. In voltaic cells the flow of electrons is spontaneous.
II. In electrolytic cells electrons flow in the external circuit (through the wire) from the anode to the cathode.
III. In voltaic cells the cathode is the positive electrode.

A. I **B.** I and III **C.** II and III **D.** I, II and III

____ **18.** Calculate ΔG for the reaction of the cell below under the stated conditions:
$Zn \mid Zn^{2+}(1.0 \times 10^{-8}\ M) \parallel Cu^{2+}(1.0 \times 10^{-6}\ M) \mid Cu$
$E°(Cu^{2+}/Cu) = +0.34$ V, $E°(Zn^{2+}/Zn) = -0.76$ V

A. -163 kJ **B.** -192 kJ **C.** 201 kJ **D.** -224 kJ

____ **19.** Which of the following cells is a concentration cell?

A. $Zn \mid Zn^{2+} \parallel Cu^{2+} \mid Cu$ **B.** $Cu \mid Cu^{2+}(0.1\ M) \parallel Cu^{2+}(1\ M) \mid Cu$

C. $Ag \mid Ag^{+} \parallel Cu^{2+} \mid Cu$ **D.** $Fe \mid Fe^{2+} \parallel Fe^{3+} \mid Fe^{2}$

____ **20.** For the cell described in question (19), EMF at 298 K is:

A. 0.118 V **B.** 0.059 V **C.** 0.0296 V **D.** 0 V

____ **21.** Why is standard reduction potential not multiplied by stoichiometric coefficients?

A. Because electrons cancel
B. Because potential is an intensive property
C. Because Gibbs energy is additive
D. Because equilibrium constant changes

____ **22.** Which statement is FALSE for a dead battery?

A. $\Delta G = 0$ **B.** $E_{cell} = 0$

C. Reaction quotient Q = K **D.** $\Delta G° = 0$

____ **23.** Calculate volume of H2 at STP after passing 1/2 F through water:

A. 11.2 L **B.** 22.4 L **C.** 5.6 L **D.** 44.8 L

____ **24.** Clinker is obtained by heating:

A. Limestone + clay + silica **B.** Limestone + gypsum

C. Clay + sand **D.** Limestone only

____ **25.** Which of the following types of glass is heat-resistant?

A. Soda-lime glass **B.** Borosilicate glass

C. Lead glass **D.** Flint glass

____ **26.** Which of the following is a primary raw material for ceramics?

A. Clay (Al2O3·2SiO2·2H2O) **B.** Na2CO3

C. Limestone **D.** CaO

____ **27.** Which by-product of sugar industry is used as fermentation substrate?

A. Bagasse **B.** Molasses **C.** Lime **D.** Gypsum

____ **28.** Synthetic detergents are more effective in hard water because:
A. Do not form insoluble salts with Ca^{2+} or Mg^{2+}
B. Are acidic
C. Contain NaOH
D. Have higher density

____ **29.** Which reaction regenerates ammonia in the Solvay process?
A. $CaCO_3 \rightarrow CaO + CO_2$ **B.** $NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$
C. $2NH_4Cl + Ca(OH)_2 \rightarrow 2NH_3 + CaCl_2 + 2H_2O$ **D.** $NH_3 + H_2O \rightarrow NH_4OH$

____ **30.** Which reaction represents the final step to obtain concentrated H_2SO_4?
A. $SO_3 + H_2O \rightarrow H_2SO_4$ **B.** $H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$
C. $SO_2 + H_2O \rightarrow H_2SO_4$ **D.** $H_2SO_4 \rightarrow H_2O + SO_3$

____ **31.** Which of the following is NOT a pesticide?
A. DDT **B.** Malathion **C.** 2,4-D **D.** Urea

____ **32.** Which reaction demonstrates the dehydrating nature of H_2SO_4?
A. $Cu + H_2SO_4 \rightarrow CuSO_4$ **B.** $C_{12}H_{22}O_{11} \rightarrow 12C + 11H_2O$
C. $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$ **D.** $SO_3 + H_2O \rightarrow H_2SO_4$

____ **33.** Which of the following pulping methods gives the highest yield?
A. Kraft pulping **B.** Sulfite pulping
C. Mechanical pulping **D.** Semi-chemical pulping

____ **34.** Which preservation method removes moisture by sublimation?
A. Drying **B.** Freeze-drying **C.** Smoking **D.** Canning

____ **35.** A 0.1 M solution of HA ($K_a = 1 \times 10^{-7}$). Its pH is approximately:
A. 2.0 **B.** 2.5 **C.** 3.0 **D.** 4.0

____ **36.** In the reaction: $HCO_3^- + H_2O \rightleftharpoons CO_3^{2-} + H_3O^+$, the conjugate acid-base pairs are:
A. HCO_3^- / CO_3^{2-} and H_3O^+ / H_2O **B.** HCO_3^- / H_2O and CO_3^{2-} / H_3O^+
C. H_2O / HCO_3^- and CO_3^{2-} / H_3O^+ **D.** HCO_3^- / H_3O^+ only

____ **37.** When soap is added to oily dirt in water, cleansing occurs mainly because:
A. Soap reduces the surface tension of water permanently
B. Soap molecules form micelles that trap oil droplets
C. Soap reacts chemically with dirt to decompose it
D. Soap increases the solubility of oil in water

____ **38.** Which property distinguishes chemical industries from other manufacturing industries?
A. Seasonal production
B. Transformation of substances through chemical reactions
C. Use of manual labor
D. Production of perishable goods

____ **39.** Number of electrons involved in the electrodeposition of 64 g of Cu from a solution of $CuSO_4$:
A. 6.02×10^{23} **B.** 3.01×10^{23} **C.** 1.204×10^{24} **D.** 6.02×10^{22}

____ **40.** On the basis of information available from the reaction: $4/3\ Al + O_2 \rightarrow 2/3\ Al_2O_3$, $\Delta G = -827$ kJ, the minimal emf required to carry out electrolysis of Al_2O_3 is:
A. 6.42 V **B.** 8.56 V **C.** 2.14 V **D.** 4.28 V